



An Interpretable Framework for Selecting Genes Relevant to Gene Ontology Biological Processes

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Abstract

High-dimensional genomic datasets present major challenges for identifying biologically meaningful features that enhance predictive modeling while remaining interpretable. This paper proposes an interpretable framework, Gram-Schmidt Feature Selection (GFS), to identify informative and minimally redundant genes through orthogonalization. The framework is evaluated on multiple benchmark gene expression datasets and compared with established feature selection methods. Results show that GFS consistently improves classification performance while reducing feature redundancy, supporting clearer model interpretation. The biological relevance of selected genes is assessed through Gene Ontology (GO) enrichment analysis, revealing significant associations with pathways and functions consistent with known cellular mechanisms. Overall, the framework demonstrates that GFS can effectively integrate predictive power with biological interpretability, offering a practical tool for selecting GO-informed genes in genomic research.

Key words: Principal Component Analysis, Gram-Schmidt Feature Selection, Interpretable Dimensionality Reduction, Gene Ontology

Introduction

Principal Component Analysis (PCA) is one of the most established and widely used techniques for dimensionality reduction. Since its development in the early 20th century, PCA has become a fundamental tool in statistical learning, pattern recognition, and data analysis [1, 9]. By identifying orthogonal directions of maximal variance and projecting the data onto a lower-dimensional linear subspace spanned by these principal components, PCA facilitates efficient representation of high-dimensional datasets. PCA not only improves computational tractability but also enhances interpretability in downstream analytical tasks.

In genomics, where datasets often comprise tens of thousands of molecular features, including gene expression levels, single nucleotide polymorphisms (SNPs), and epigenetic markers, PCA has become an indispensable exploratory tool [20]. A well-established application arises in genome-wide association studies (GWAS), in which PCA is routinely employed to detect and correct for population stratification, thereby mitigating confounding effects attributable to ancestry-related structure [19, 16]. In transcriptomic research, PCA is widely used to extract global expression patterns, reduce technical variation,

and reveal biologically meaningful sample groupings, such as tumor subtypes and differentiation states [3]. The rapid development of single-cell transcriptomic technologies has further underscored the importance of PCA, as it provides a key initial step for capturing dominant axes of cellular heterogeneity and facilitates the visualization of distinct cell populations and developmental trajectories. In epigenomic and proteomic studies, PCA similarly enables the identification of major sources of variation that correspond to interpretable biological processes, including cell cycle progression, tissue specificity, and responses to environmental perturbations [12].

Beyond visualization and stratification control, PCA facilitates downstream analyses by reducing the dimensionality of omics datasets, which alleviates the curse of dimensionality and improves the performance and stability of statistical models [8]. Principal components are frequently incorporated as covariates in regression frameworks to account for latent sources of heterogeneity, or used as input features for clustering algorithms to enhance robustness and computational tractability. Despite its broad utility, a major limitation of PCA arises from its linearity assumption, which is often incompatible with the complex and nonlinear behavior of biological systems. Genomic regulation involves high-order gene gene interaction networks, epigenetic modifications that operate

as molecular switches, and feedback mechanisms that generate dynamic and non-proportional responses [2]. Furthermore, in single-cell transcriptomic data, the substantial sparsity introduced by dropout events constrains the effectiveness of PCA, because linear projections struggle to capture relationships within zero-inflated and highly heterogeneous expression distributions. As a result, reliance solely on PCA may yield oversimplified or distorted representations of the underlying biological structure [6, 8].

Despite its widespread use and strong performance in many settings, PCA exhibits several important limitations that restrict its applicability in modern biological data analysis. First, PCA is inherently a linear method, and it therefore cannot capture nonlinear relationships that often characterize gene regulatory networks, cellular signaling pathways, and other complex biological processes [22, 23, 21]. Second, PCA performs poorly in high dimensional settings in which the number of variables greatly exceeds the number of samples, because the sample covariance matrix becomes unstable and the leading principal components may be dominated by noise rather than true biological signal [7, 17, 9]. Third, PCA struggles with highly sparse datasets, such as single cell transcriptomic data, where dropout events and zero inflation distort variance estimates and lead linear projections to misrepresent relationships among samples [18, 10, 14]. These limitations highlight the need for more flexible dimensionality reduction methods that account for nonlinear structure, sparsity, and high dimensional statistical behavior.

To address these limitations, we propose the Gram-Schmidt Feature Selection (GFS) algorithm. GFS is an unsupervised feature selection algorithm that iteratively identifies and removes redundant features using a Gram-Schmidt orthogonalization over chosen multilinear polynomials. At each step, GFS computes a variance vector for the current transformed data, selects the feature with the largest remaining variance, and then orthogonalizes the function family with respect to this newly selected feature so that nonlinear dependencies can be removed before the next step. The process continues until the maximum remaining variance drops below a user-defined threshold, yielding a subset of features that retains the informative structure of the data while discarding those that are linearly or nonlinearly redundant. GFS is used as an add-on unit before applying PCA for downstream analyses. It is a feature selection method capable of capturing nonlinear dependencies between features, because it applies a Gram-Schmidt orthogonalization over multilinear polynomials. This is particularly relevant in many genomics applications, where features are often binary (e.g., mutation presence/absence, motif matches), and where multilinear polynomials naturally and expressively represent complex nonlinear interactions among such variables. It operates effectively even on sparse datasets, since GFS relies only on expectations of $\mathcal{L}^2$ functions, quantities that remain well-defined and stable for low-degree multilinear polynomials (e.g., degree 2) even under heavy sparsity. Finally, it is well-suited for high-dimensional settings, as its iterative evaluation of individual features allows it to prune redundant dimensions early, reducing computational load, and its structure scales primarily with the size of the selected function family rather than the full data dimension, making it efficient and practical for large-scale genomic and other high-dimensional analyses.

Finally, to validate our theoretical results, we conduct extensive numerical experiments on benchmark datasets. To demonstrate that GFS selects biologically meaningful genes, we apply GFS

to the 10x dataset, which is a well-known dataset in spatial genomics [15]. We then perform PCA on the selected features and compare the resulting gene ontology analysis with the analysis obtained by applying PCA to the full dataset. Our results indicate that several features selected by PCA on the full dataset are irrelevant to the underlying biological system, specifically the Molecular, Spatial, and Functional Single-Cell Profiling of the Hypothalamic Preoptic Region dataset. To further evaluate whether this improved informativeness leads to better classification performance, we apply kNN to both scenarios and show that the results obtained after applying GFS outperform those obtained without GFS. Finally, we compare GFS with several well-known feature selection methods in the literature and show that its classification accuracy is significantly higher than that of existing algorithms.

The rest of the paper is organized as follows: We first introduce the Gram-Schmidt Feature Selection (GFS) method and describe the dataset used to evaluate our proposed algorithm. Subsequently, we present the experiments conducted, outlining our evaluation methodology which assesses performance in terms of classification accuracy, gene ontology analysis, and the relevance of the selected genes to the benchmark dataset. Finally, the paper discusses the results obtained, including comparisons with existing algorithms, and provides a comprehensive analysis of the selected features' biological significance.

Method

In this section, we first introduce the Gram-Schmidt Feature Selection (GFS) method. We then describe the dataset used to evaluate our proposed algorithm, followed by a presentation of the experiments conducted. Finally, we outline our evaluation methodology, which assesses performance in terms of classification accuracy, gene ontology analysis, and the relevance of the selected genes to the benchmark dataset.

Proposed Algorithm: GFS

Notation

Let $X = [X_1, \dots, X_d]^\top \in \mathbb{R}^d$ denote a random feature vector whose entries are real-valued and sampled i.i.d. from an unknown distribution P_X . Deterministic vectors are written as $\mathbf{x} = [x_1, \dots, x_d]^\top$. We use $[d] = \{1, \dots, d\}$ and, for $j \leq d$, set $s_{[j]} = \{s_1, \dots, s_j\}$. For any subset $S \subseteq [d]$, the notation X_S (resp. $\mathbf{x}_S$) refers to a vector of length $|S|$ which contains the entries of X (resp. $\mathbf{x}$) indexed by S ; likewise, $X_{s_{[j]}}$ collects the entries of X in $s_{[j]}$. Let $\mathcal{P}(\mathbf{x})$ be a family of multilinear polynomials in $\{x_1, \dots, x_d\}$. For $S \subseteq [d]$, the set $\mathcal{P}(\mathbf{x}_S)$ consists of those polynomials depending only on variables with indices in S . When replacing x_i by their random counterparts, we write $\mathcal{P}(X_S)$ for the result of substituting $x_i \leftarrow X_{s_i}$ for $S = \{s_1 < \dots < s_{|S|}\}$. We use the algorithmic notation $a \leftarrow b$ to denote that the expression b is computed, and then assigned into the variable a .

Definition 1 A multilinear polynomial is a polynomial in which each variable appears with degree at most 1; equivalently, it is linear in every variable when all other variables are held constant. For example $f(a, b, c) = 3a + 2bc - 5ac$ is multilinear because no variable appears with power greater than 1 in any term.

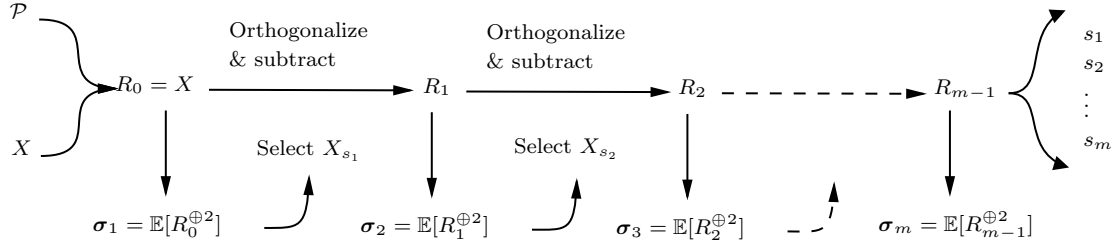


Fig. 1: **A schematic illustration of GFS.** The variance vectors $\{\sigma_i\}_{i=1}^m$ are computed iteratively by subtracting projections onto the already orthogonalized functions in $\mathcal{P}$, whose variables X_{s_i} have been fixed as the selected features of X . Each iteration yields a new random variable R_j together with its corresponding variance vector σ_j . These variance vectors subsequently determine, at every step, the feature of R_j with the largest variance, which is then selected for the next iteration.

Example 1 Let $d = 4$ and $\mathcal{P}(\mathbf{x}) = \{x_1, x_2, x_4, x_1x_4, x_2x_3x_4\}$. For $\mathcal{S} = \{1, 4\}$ we have $\mathcal{P}(\mathbf{x}_{\mathcal{S}}) = \{x_1, x_4, x_1x_4\}$, and for $\mathcal{T} = \{2, 3\}$ we obtain $\mathcal{P}(\mathbf{x}_{\mathcal{T}}) = \{x_2, x_3\}$.

Our methods do not require any specific assumptions on $\mathcal{P}(\mathbf{x})$ or on P_X , except that $\mathcal{P}(X_{\mathcal{S}}) \subseteq \mathcal{L}^2(P_X)$ for every $\mathcal{S} \subseteq [d]$. In other words, after substituting x_i with X_i , each resulting function must have finite variance and therefore lie in the Hilbert space equipped with the inner product $\langle f, g \rangle = \mathbb{E}[fg]$. Throughout the paper, every random quantity is a measurable function of the base random vector X , from which the data are drawn i.i.d., and all expectations are taken with respect to P_X .

Remark 1 The random vector X used in this paper does not need to be explicitly specified. We manipulate random variables symbolically and rely on them only through expectations. In practice, these expectations are approximated by empirical means. For example, one may form a function $g(X) = X_1 + X_2^2$ without needing to know the distribution of X . To keep the notation simple, we write expressions such as $\mathbb{E}[g(X)]$ instead of the empirical approximation $\mathbb{E}[g(X)] \approx \sum_{i=1}^N g(\alpha_i) = \sum_{i=1}^N (\alpha_{i,1} + \alpha_{i,2}^2)$, where $\{\alpha_i\}_{i=1}^N$ is the dataset.

Gram-Schmidt Orthogonalization over Function Spaces

In this section, we explain the Gram-Schmidt (GS) process on function spaces. We denote by $\hat{\mathcal{P}}(X_{\mathcal{S}})$ the result of applying the Gram-Schmidt process over multilinear polynomials $\mathcal{P}(X_{\mathcal{S}})$. Throughout this paper, we make use of the well known Hadamard product, denoted by $\oplus$, where for two vectors $\mathbf{x} = (x_i)_{i=1}^n, \mathbf{y} = (y_i)_{i=1}^n$ we have $\mathbf{x} \oplus \mathbf{y} = (x_i y_i)_{i=1}^n$; similarly $\mathbf{x}^{\oplus 2} = (x_i^2)_{i=1}^n$.

We fix multilinear polynomials $\mathcal{P}(\mathbf{x})$ in variables $\mathbf{x} = (x_1, \dots, x_d)$. Our algorithm proceeds in an iterative manner: at iteration j , we select a new feature X_{s_j} , apply the substitution $x_j \leftarrow X_{s_j}$ to the functions in $\mathcal{P}(\mathbf{x}_{[j]})$, and then perform GS orthogonalization. At each iteration j , Algorithm 1 starts with the previously orthogonalized multilinear polynomials $\hat{\mathcal{P}}(X_{s_1}, \dots, X_{s_{j-1}})$, together with the newly introduced functions in $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$ and the new random variable X_{s_j} . For every function in $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$, the algorithm substitutes $x_i \leftarrow X_{s_i}$ for all $i \in [j]$ (Line 5), subtracts its projection onto the functions already orthogonalized (Line 6), normalizes the resulting function (Line 7), and appends it to $\mathcal{T}$ (Line 8). After this projection step, the algorithm defines a new random vector R_j derived from X (Line 11), computes its variance vector and returns it as output (Line 12).

Algorithm 1 $\text{GS}(\hat{\mathcal{P}}(X_{s_{[j-1]}}, \mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]}), X_{s_{[j]}})$

Require:

- $\hat{\mathcal{P}}(X_{s_{[j-1]}})$: orthogonalized multilinear polynomials.
- $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]})$: all the multilinear polynomials in $\mathcal{P}(\mathbf{x})$ which depend on $\mathbf{x}_{[j]}$ but not only on $\mathbf{x}_{[j-1]}$.
- $X_{s_{[j]}}$: the random variables which appear in $\hat{\mathcal{P}}(X_{s_{[j-1]}})$, and a new random variable $X_{s_{[j]}}$.

Ensure: Orthogonalized multilinear polynomials $\hat{\mathcal{P}}(X_{s_{[j]}})$ and a variance vector σ_{j+1} .

- 1: **Initialize:** $\mathcal{T} \leftarrow \hat{\mathcal{P}}(X_{s_{[j-1]}})$.
 - 2: Denote $\mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]}) \triangleq \{f_1(\mathbf{x}), \dots, f_{\ell}(\mathbf{x})\}$,
 - 3: where $f_1(\mathbf{x}) = x_j$.
 - 4: **for** $k \leftarrow 1$ **to** ℓ **do**
 - 5: $g(X_{s_{[j]}}) \leftarrow f_k(X_{s_{[j]}})$
 - 6: $g(X_{s_{[j]}}) \leftarrow g(X_{s_{[j]}}) - \sum_{f \in \mathcal{T}} \mathbb{E}[gf]f$
 - 7: $g(X_{s_{[j]}}) \leftarrow \frac{g(X_{s_{[j]}})}{\sqrt{\mathbb{E}[g(X_{s_{[j]}})^2]}}$
 - 8: $\mathcal{T} \leftarrow \mathcal{T} \cup \{g(X_{s_{[j]}})\}$
 - 9: **end for**
 - 10: Define $\hat{\mathcal{P}}(X_{s_{[j]}}) = \mathcal{T}$.
 - 11: Define $R_j = X - \sum_{f \in \mathcal{T}} \mathbb{E}[Xf]f$.
 - 12: Define $\sigma_{j+1} = \mathbb{E}[R_j^2]$.
 - 13: Return $\sigma_{j+1}, \mathcal{T}$.
-

Gram-Schmidt Feature Selection (GFS)

Algorithm 2 presents the *Gram-Schmidt Feature Selection* (GFS) method. At iteration j , the algorithm examines the variance vector σ_j to determine which feature of R_{j-1} exhibits the largest variance. If this maximal variance is below the threshold ϵ^2 , the procedure halts. Otherwise, the feature of R_{j-1} with highest variance is selected and denoted by X_{s_j} . The algorithm then performs a Gram-Schmidt orthogonalization step on the multilinear functions that involve $X_{s_{[j]}}$ but exclude any dependence only on $X_{s_{[j-1]}}$.

Dataset

In this study, we incorporated the scRNA-seq of the preoptic region in the mouse hypothalamus [15], which provides a comprehensive molecular, spatial, and functional survey of the hypothalamic preoptic region in *Mus musculus*. This dataset was generated by Moffitt *et al.* using high-throughput scRNA-seq on the Illumina NextSeq 500 platform, profiling approximately

Algorithm 2 Gram-Schmidt Feature Selection (GFS)

Require: Multilinear polynomials $\mathcal{P}(\mathbf{x})$ and a threshold $\epsilon > 0$.
Ensure: Selected features $s_1, \dots, s_m$ (m is a varying number that depends on $\mathcal{P}(\mathbf{x})$, ϵ , and X)

```

1: Initialize:  $\sigma_1 = \mathbb{E}[X^{\oplus 2}]$ .
2: for  $j \leftarrow 1$  to  $d$  do
3:   Let  $s_j \triangleq \arg \max_{i \in [d]} \{\sigma_{j,i}\}_{i=1}^d$ , where  $\sigma_j = (\sigma_{j,i})_{i=1}^d$ .
4:   if  $\|\sigma_j\|_\infty \leq \epsilon^2$  then
5:     break.
6:   end if
7:    $\sigma_{j+1}, \hat{P}(X_{s_{[j]}}) = \text{GS}(\hat{P}(X_{s_{[j-1]}}), \mathcal{P}(\mathbf{x}_{[j]}) \setminus \mathcal{P}(\mathbf{x}_{[j-1]}), X_{s_{[j]}})$ 
8: end for

```

31,000 dissociated cells collected from the preoptic area of multiple adult male and female mice. The experimental design includes six samples comprising three biological replicates from pooled female mice and three biological replicates from pooled male mice, enabling robust comparisons of sex-specific transcriptional signatures. The work was produced by the Moffitt Laboratory at Boston Children’s Hospital and is publicly accessible through the Gene Expression Omnibus (GEO). Overall, this dataset offers a detailed cellular atlas of the preoptic hypothalamus and serves as a valuable resource for investigating hypothalamic cell diversity and function.

Experimental Setup

To evaluate the effectiveness of our proposed GFS method, we first applied GFS to the dataset without using any class labels, consistent with the unsupervised nature of the algorithm. To implement GFS, we used multilinear polynomials of degree 2 in $\mathcal{P}$ functions, and all components of the method were implemented in Python. After obtaining the subset of features selected by GFS, we analyzed the results from multiple perspectives, including gene ontology enrichment and visualization, classification accuracy, and comparisons with several well-known feature selection algorithms in the literature.

Gene Ontology

We conducted a comparative analysis against the full, raw dataset to assess whether the selected features by GFS capture biologically meaningful variation related to the underlying cell types. Specifically, we performed principal component analysis (PCA) on both the raw gene expression matrix and the reduced feature set produced by GFS, extracting the top 20 principal components in each condition. For each principal component, we identified 100 genes with the largest absolute coefficients, representing the features most influential in defining the major genes in each principal component. Using the PCA-derived gene sets, we categorized the features into three groups:

1. Genes shared between the PCA results of the full dataset and those of the GFS-selected subset
2. Genes uniquely identified through PCA on the full dataset
3. Genes uniquely identified through PCA on the GFS-selected subset.

We then perform gene ontology (GO) enrichment analysis using the R package Enrichr [11] on each group to determine their functional relevance and to assess whether GFS preferentially retains genes associated with known cell-type specific biological

processes. Gene set library “GO.Biological_Process_2018” is used with default parameters. The combined score is defined as the P-value computed using the Fisher’s exact test multiplied with the z-score of the deviation from the expected rank. We also use Uniform Manifold Approximation and Projection (UMAP), implemented through the `umap-learn` package, to visualize the results. we employ PCA implementation from `sklearn.decomposition` module.

To show the importance of the results of GO, we divide them into three categories as follows:

- **Relevant:** Biological processes, pathways, or molecular functions that are directly intrinsic to the system or tissue under study, as well as those that, while not unique to it, are commonly active, functionally important, or contextually significant within it. Together, these terms encompass both the core cellular mechanisms, development, and physiology of the primary cell types of interest, and the supporting or regulatory biology that directly influences their function.
- **Weakly relevant:** Processes that are general, systemic, or peripheral to the main biology of the system, but that may appear due to cross-talk, infiltration, or shared molecular machinery. They are biologically valid but not central to the dataset’s focus.
- **Irrelevant:** Processes with no clear or plausible connection to the biology of the system being studied. These terms likely arise due to background noise, annotation artifacts, or statistical enrichment unrelated to the experimental context.

Classification Accuracy

To compare the results quantitatively, we evaluated both the raw data and the features selected by GFS by running a classification task after applying PCA. Finally, to assess how well the selected features support downstream predictive tasks, we measured classification performance using a k-nearest neighbors (kNN) classifier trained on the resulting low-dimensional embeddings. This evaluation allowed us to quantify whether the unsupervised feature selection carried out by GFS preserves the discriminative structure necessary for distinguishing cell types. we employed kNN implementation from the `sklearn.neighbors` module in Python.

Comparison of GFS with Existing Feature Selection Algorithms

In addition to analyzing GFS in isolation, we compared it against several well-established feature selection algorithms from the literature—namely Multi-Cluster Feature Selection (MCFS) [4], Nonnegative Discriminative Feature Selection (NDFS) [13], Laplacian Score (LS) [5], and Spectral Feature Selection (SPEC) [24]—to assess its performance relative to existing methods, with classification accuracy serving as the primary evaluation metric. To implement MCFS, NDFS, LS, and SPEC, we used the `skfeature-chappers` package.

Results

In this section, we present the numerical simulation results of GFS, which is described in detail in Section 2. Figure 2 illustrates a detailed schematic of the first set of experiments, including the overall workflow and evaluation procedures. These experiments are designed to assess the ability of GFS to identify the most informative features from high-dimensional data.

Specifically, we report three complementary types of results: (1) classification accuracy, which evaluates the predictive performance of the selected features; (2) gene ontology analysis, which examines the biological relevance and functional enrichment of the selected genes; and (3) visualization, which provides an intuitive interpretation of the feature selection behavior of GFS.

Classification Accuracy

In this section, we compare the classification accuracy (%) of PCA applied to both the raw gene expression matrix and the reduced feature set generated by GFS. The quantitative results are summarized in Table 1. We consider three categories of data: (1) the entire dataset, comprising 31,299 cells; (2) excitatory neurons, comprising 3,511 cells; and (3) inhibitory neurons, comprising 15,042 cells. Across all three cases, the results reported in Table 1 demonstrate that applying GFS consistently improves classification accuracy. Notably, even in the setting with a relatively small number of samples—namely, the excitatory neuron subset—GFS yields strong classification performance, highlighting its effectiveness in low-sample regimes.

Table 1. Comparison of classification accuracy (%) of PCA using the raw gene expression matrix and the reduced feature set obtained via GFS for the full dataset, excitatory neurons, and inhibitory neurons.

Dataset	Entire dataset	Excitatory	Inhibitory
GFS	90.70	57.33	51.97
Raw Data	86.67	50.36	48.45

Gene Ontology Analysis

In this section, we present the results for gene ontology analysis detailed in section 2.3.1. The GO analysis highlights distinct differences between the biological processes identified by GFS.

When PCA was applied to the GFS-selected genes, the resulting gene set emphasized pathways directly related to neural and metabolic regulation. Several enriched categories were associated with amino acid and carboxylic acid transport (GO:0089718, GO:0015804, GO:0006865, GO:0046942), processes that are essential for neurotransmitter precursor uptake, glutamate/GABA cycling, and metabolic coupling between neurons and glia. Additional terms highlighted inhibitory G protein-coupled receptor (GPCR) signaling (GO:0007193) and the regulation of cell differentiation (GO:0045595), both of which play central roles in hypothalamic signal transduction and lineage specification. Enrichment of processes involved in the regulation of endopeptidase and peptidase activity (GO:0010951, GO:0052548, GO:0010466) further underscored mechanisms controlling peptide hormone activity and extracellular signaling. Collectively, these results indicate that PCA on GFS-selected genes captures pathways that reflect core aspects of hypothalamic function, including neurotransmission, cellular differentiation, and metabolic regulation.

In contrast, when PCA was applied to the entire gene expression dataset, the resulting gene set exhibited a strong bias toward immune-related categories. Enriched terms included antimicrobial humoral immune responses (GO:0061844, GO:0019730), macrophage chemotaxis and migration (GO:0048246, GO:1905517), and neutrophil or granulocyte activation and

chemotaxis (GO:0042119, GO:0071621). Although these pathways may reflect microglial activity or immune infiltration signatures, they are not central to hypothalamic biology. Moreover, several enriched categories, such as regulation of osteoblast proliferation (GO:0033688), were biologically irrelevant to neural tissue, suggesting spurious enrichment. Only a limited subset of terms, including cellular responses to oxygen-containing compounds (GO:1901701), showed partial relevance to neural or metabolic processes.

Taken together, these findings demonstrate that GFS identifies gene sets that map to biologically coherent pathways central to hypothalamic and neuronal function. This reinforces the advantage of GFS in prioritizing functionally meaningful features in complex single-cell datasets.

Visualization

In this subsection, we investigate the biological interpretability of the selected features through gene ontology analysis. By comparing PCA applied to GFS-selected genes with PCA applied to the entire gene expression dataset, we assess how feature selection influences the functional relevance of the resulting gene sets. This analysis provides insight into the ability of GFS to prioritize biologically meaningful signals while suppressing spurious or non-informative patterns. Figure 4 illustrates how GFS improves visualization, enabling clearer distinction among different cell types.

GFS vs other feature selection algorithms

Figure 5 presents a comparative evaluation of the proposed algorithm, GFS, against several established feature selection methods, including Multi-Cluster Feature Selection (MCFS), Nonnegative Discriminative Feature Selection (NDFS), Laplacian Score (LS), and Spectral Feature Selection (SPEC). The results demonstrate that GFS consistently achieves higher classification accuracy across varying numbers of selected features, indicating its strong performance on genomic datasets. Although MCFS, NDFS, LS, and SPEC exhibit incremental improvements as the number of selected features increases, their performance remains consistently inferior to that of GFS throughout the evaluation. This sustained advantage highlights GFS's ability to more effectively identify discriminative and informative features, reduce redundancy, and improve generalization relative to existing methods. Overall, these results confirm the effectiveness of GFS and underscore its potential as a robust feature selection strategy for high-dimensional learning tasks, particularly in sparse data settings.

Conclusion

This paper introduces the Gram-Schmidt Feature Selection (GFS) method as a powerful tool for identifying informative genes from high-dimensional genomic data. Our extensive evaluation demonstrates that GFS consistently outperforms traditional feature selection methods by improving classification accuracy and reducing redundancy. The method's effectiveness is highlighted across various datasets and experimental settings, including low-sample regimes. Additionally, gene ontology analysis confirms the biological relevance and functional enrichment of the selected genes, further validating the utility of GFS in genomic studies. These results underscore the potential of GFS to enhance our

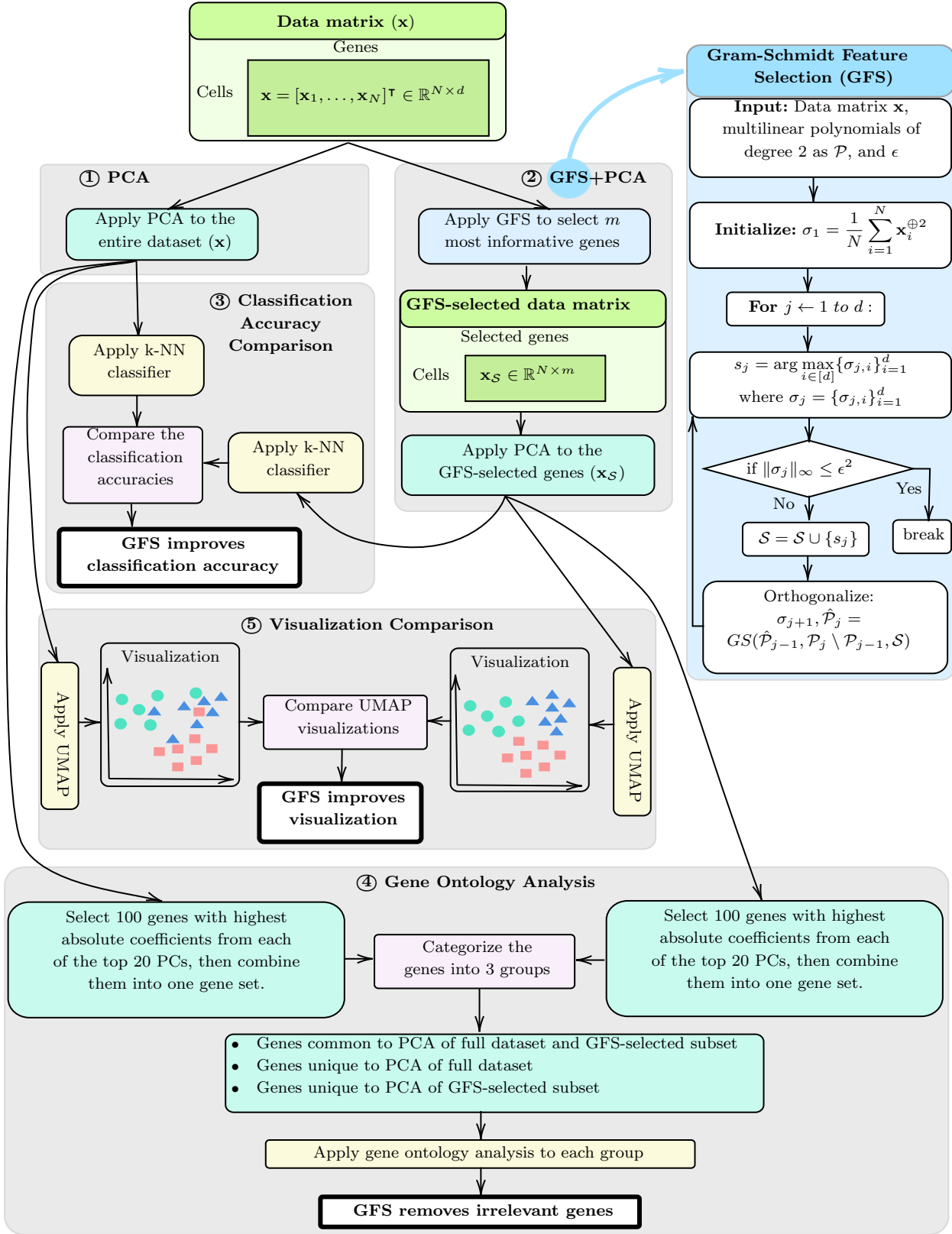


Fig. 2: A schematic illustration of the experimental setup.

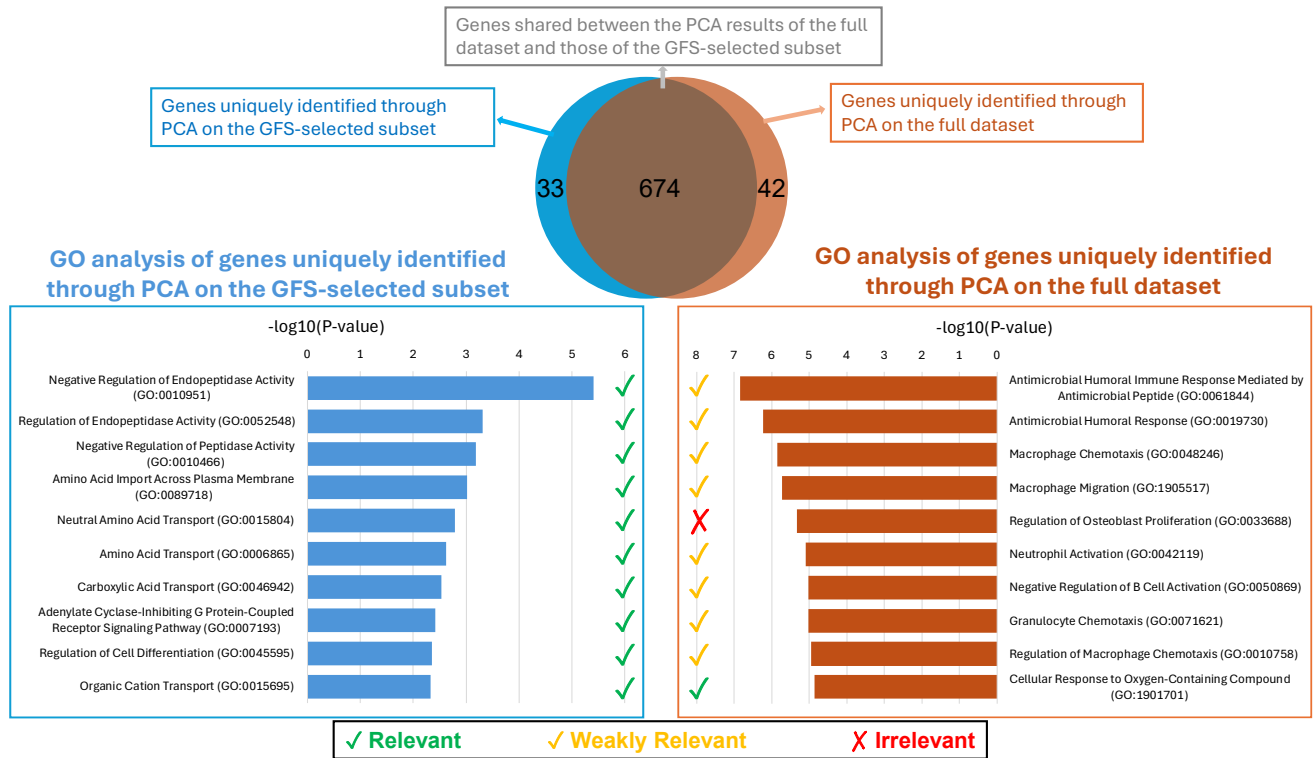


Fig. 3: Gene ontology enrichment analysis comparing PCA applied to GFS-selected genes and PCA applied to the full gene expression dataset. PCA on GFS-selected genes highlights pathways related to neural signaling, cellular differentiation, and metabolic regulation, whereas PCA on the full dataset yields predominantly immune-related and biologically irrelevant categories, illustrating the improved functional relevance achieved through GFS-based feature selection.

understanding of gene regulation and cellular functions, making it a valuable resource for both research and clinical applications.

Competing interests

The authors declare that they have no competing interests.

Author contributions statement

B.Y. conceptualized the study, curated the data, performed the formal analysis, carried out the investigation, developed the methodology, administered the project, provided resources, developed the software, validated the findings, generated the visualizations, drafted the original manuscript, and wrote and reviewed the final version. B.S. developed the methodology, provided funding, administered the project, provided resources, supervised the work, and wrote and reviewed the final version. Y.L. conceptualized the study, curated the data, performed the formal analysis, carried out the investigation, developed the methodology, provided resources, supervised the work, validated the findings, drafted the original manuscript, and wrote and reviewed the final version.

References

- Hervé Abdi and Lynne J Williams. Principal component analysis. *Wiley interdisciplinary reviews: computational statistics*, 2(4):433–459, 2010.
- Uri Alon. *An introduction to systems biology: design principles of biological circuits*. Chapman and Hall/CRC, 2019.
- Orly Alter, Patrick O Brown, and David Botstein. Singular value decomposition for genome-wide expression data processing and modeling. *Proceedings of the National Academy of Sciences*, 97(18):10101–10106, 2000.
- Deng Cai, Chiyuan Zhang, and Xiaofei He. Unsupervised feature selection for multi-cluster data. In *Proceedings of the 16th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 333–342, 2010.
- Xiaofei He, Deng Cai, and Partha Niyogi. Laplacian score for feature selection. *Advances in neural information processing systems*, 18, 2005.
- Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An introduction to statistical learning: with applications in R*, volume 103. Springer, 2013.
- Iain M Johnstone. On the distribution of the largest eigenvalue in principal components analysis. *The Annals of statistics*, 29(2):295–327, 2001.
- Iain M Johnstone and D Michael Titterton. Statistical challenges of high-dimensional data, 2009.
- Ian Jolliffe. Principal component analysis. In *International encyclopedia of statistical science*, pages 1094–1096. Springer, 2011.
- Dmitry Kobak and Philipp Berens. The art of using t-sne for single-cell transcriptomics. *Nature communications*, 10(1):5416, 2019.

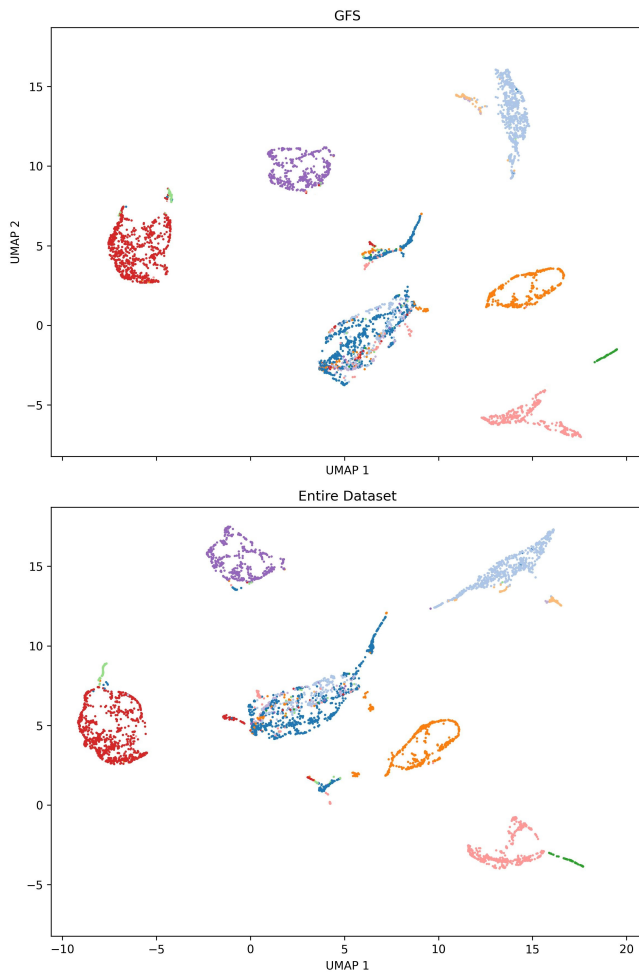


Fig. 4: UMAP visualization comparing GFS-selected genes with the full gene expression dataset.

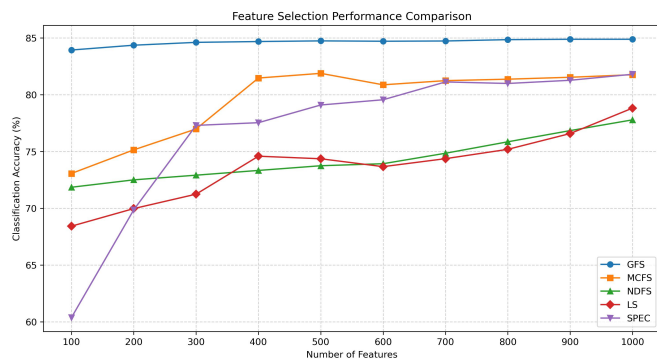


Fig. 5: Comparison of feature selection methods (GFS, MCFS, NDFS, LS, and SPEC) across varying numbers of selected features.

11. Ma'ayan Lab. Enrichr: Interactive and collaborative enrichment analysis. <https://maayanlab.cloud/Enrichr/>, 2025. Accessed: 2025-02-01.
12. Jeffrey T Leek, Robert B Scharpf, Héctor Corrada Bravo, David Simcha, Benjamin Langmead, W Evan Johnson, Donald

- Geman, Keith Baggerly, and Rafael A Irizarry. Tackling the widespread and critical impact of batch effects in high-throughput data. *Nature Reviews Genetics*, 11(10):733–739, 2010.
13. Zechao Li, Yi Yang, Jing Liu, Xiaofang Zhou, and Hanqing Lu. Unsupervised feature selection using nonnegative spectral analysis. In *Proceedings of the AAAI conference on artificial intelligence*, volume 26, pages 1026–1032, 2012.
14. Malte D Luecken and Fabian J Theis. Current best practices in single-cell rna-seq analysis: a tutorial. *Molecular systems biology*, 15(6):e8746, 2019.
15. Jeffrey R Moffitt, Dhananjay Bambah-Mukku, Stephen W Eichhorn, Eric Vaughn, Karthik Shekhar, Julio D Perez, Nimrod D Rubinstein, Junjie Hao, Aviv Regev, Catherine Dulac, et al. Molecular, spatial, and functional single-cell profiling of the hypothalamic preoptic region. *Science*, 362(6416):eaau5324, 2018.
16. Nick Patterson, Alkes L Price, and David Reich. Population structure and eigenanalysis. *PLoS genetics*, 2(12):e190, 2006.
17. Debashis Paul. Asymptotics of sample eigenstructure for a large dimensional spiked covariance model. *Statistica Sinica*, pages 1617–1642, 2007.
18. Emma Pierson and Christopher Yau. Zifa: Dimensionality reduction for zero-inflated single-cell gene expression analysis. *Genome biology*, 16(1):241, 2015.
19. Alkes L Price, Nick J Patterson, Robert M Plenge, Michael E Weinblatt, Nancy A Shadick, and David Reich. Principal components analysis corrects for stratification in genome-wide association studies. *Nature genetics*, 38(8):904–909, 2006.
20. Markus Ringnér. What is principal component analysis? *Nature biotechnology*, 26(3):303–304, 2008.
21. Sam T Roweis and Lawrence K Saul. Nonlinear dimensionality reduction by locally linear embedding. *science*, 290(5500):2323–2326, 2000.
22. Bernhard Schölkopf, Alexander Smola, and Klaus-Robert Müller. Kernel principal component analysis. In *International conference on artificial neural networks*, pages 583–588. Springer, 1997.
23. Joshua B Tenenbaum, Vin de Silva, and John C Langford. A global geometric framework for nonlinear dimensionality reduction. *science*, 290(5500):2319–2323, 2000.
24. Zheng Zhao, Lei Wang, and Huan Liu. Efficient spectral feature selection with minimum redundancy. In *Proceedings of the AAAI conference on artificial intelligence*, volume 24, pages 673–678, 2010.

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